Practical Machine Learning Insights: Regression Tips

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Difference Between Regression and Classification

- Regression: Used when the goal is to predict a continuous real value, such as predicting a salary, a house price, or temperature.

- Classification: Used when the goal is to predict a discrete category or a specific class, such as (Yes OR No), (Cat OR Dog) or (spam OR not spam).

Feature Scaling in Multiple Linear Regression

- In a standard Multiple Linear Regression model (using Ordinary Least Squares), explicit feature scaling (like Standardization or Normalization) is not required.

  - **The Logic:** The model's equation includes a specific coefficient $(b_0, b_1, b_2, \dots \dots, b_n)$ for each independent feature. If one feature is measured in millions and another in small decimals, the algorithm will automatically compensate by assigning a proportionally smaller coefficient to the larger scale and a larger coefficient to the smaller scale.

  - **The Result:** The final prediction remains identical whether you scale the data or not.

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Dummy Variable Trap

- When using the Scikit-Learn **LinearRegression** class, you don’t need to manually drop one dummy column. The library’s underlying solver is robust enough to handle multicollinearity automatically, preventing the '**trap**' from breaking the model.

Backward Elimination for Feature Selection

- While modern libraries will process all provided features without crashing, they do not automatically perform Backward Elimination. However, manual elimination is often unnecessary for prediction accuracy; the model will naturally assign near-zero coefficients to insignificant features.

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